

Christian Scaff

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Education

Columbia University

M.S. in Computer Science, personalized track (expected Dec 2026)

Advised by Prof. Mark Santolucito

Sep 2025 – Dec 2026

GPA: 3.62/4.0

University of Florida

B.S. in Computer Science

Aug 2021 – May 2025

GPA: 3.96/4.0

Publications

PALS: Preference-guided Active Automata Learning for Symbolic Reinforcement Learning in Games

William Fishell, **Christian Scaff**, Sam Kouteili, and Mark Santolucito. *NExT-Game: New Frontiers in Game-Theoretic Learning*, ICML 2026 Workshop

openreview.net/forum?id=jRUJpASPQJ

Mining Beyond the Booleans: Learning Data Transformations and Temporal Specifications

Sam Kouteili, William Fishell, **Christian Scaff**, Mark Santolucito, and Ruzica Piskac. To appear, *ESOP 2027*

arxiv.org/abs/2603.06710

Experience

Research Lead

Columbia University – Formal Methods and Reasoning Group (FoRG)

Sep 2025 – Present

New York, NY

- Recruited and now lead a ten-person team across developer experience, systems, and hardware, executing the FPGA research agenda I first-authored, which targets a public beta in December 2026. manhattanreasoning.com/research
- Led development of the cloud FPGA platform with two other students: seven Lattice ECP5-85F boards running LiteX/VexRiscv SoCs, a FastAPI and Redis orchestrator, and on-demand AWS Fargate build tasks.
- Published *manhattan-reasoning-gym* to PyPI: containerized Yosys and nextpnr synthesis, sandboxed agent execution, and remote FPGA programming.

Entrepreneurial Lead

NSF I-Corps National Program (\$50,000 award) – Manhattan Reasoning

Jun 2026 – Aug 2026

New York, NY

- Ran 100 customer interviews with hardware and EDA engineers, AI researchers, and quant firms for a venture commercializing verifiable RL environments for RTL design.
- Recommended against commercializing on a lack of product-market fit and transitioned the venture into an open-source research project.

Research Intern

Georgia Tech Research Institute

May 2025 – Aug 2025

Atlanta, GA

- Developed Wy-Graph, a Python tool that translates LLVM IR into knowledge graphs through 28 node types and 41 edge types, producing training data for differentiable inductive logic programming over program structure.

Research Programmer

University of Florida GATAS Laboratory

Aug 2024 – May 2025

Gainesville, FL

- Wrote four domain-specific languages for AlgebraicJulia, an open-source Julia ecosystem for applied category theory, converting three handwritten DSLs (ACSets.jl, Catlab.jl, DiagrammaticEquations.jl) into parsing expression grammars.
- Implemented a DSL for cellular sheaf creation in AlgebraicOptimization.jl with MLStyle.jl.

Technology Architecture Analyst

Accenture

Jun 2024 – Aug 2024

Chicago, IL

- Replaced a manual monthly IT reporting process for a healthcare client with an automated Power BI report.
- Analyzed five months of ServiceNow CMDB data and consolidated it into an interactive Power BI dashboard.

Independent Research

Equivalence Checking LLM-Optimized GPU Code

Columbia University – COMS E6901, advised by Prof. Stephen A. Edwards

Spring 2026

New York, NY

- Tested whether LLM-generated CUDA optimizations can be proven to preserve the semantics of the C++ they replace, for the equivalence checker in the front-end of MACCHIATO, a compiler toolchain under DARPA's MOCHA program.
- Found that model checking with CBMC does not scale to real CUDA kernels, and recommended an agent-driven Lean 4 theorem prover approach instead.

Teaching

Undergraduate Teaching Assistant
University of Florida CISE Department

Jan 2024 – Dec 2024
Gainesville, FL

- Held weekly office hours for COP 4020: Programming Language Concepts, a course on programming language design and implementation.

Open Source Contributions

Author & Maintainer

- **Manhattan Reasoning Gym** – FPGA synthesis and agent-sandboxing toolkit, published on PyPI. pypi.org/project/manhattan-reasoning-gym
- **Wyvern Graph** – LLVM IR to knowledge graph translator, released under MIT. github.com/cscaff/Wyvern-Graph

Contributor

- **AlgebraicJulia** – Applied category theory in Julia; contributed to ACSets.jl, AlgebraicOptimization.jl, Catlab.jl, and DiagrammaticEquations.jl. github.com/AlgebraicJulia

Skills

Languages	Python, C/C++, Julia, Java, TypeScript, SQL (MariaDB)
Hardware Design	SystemVerilog, Amaranth HDL, SystemC, LiteX, VexRiscv
EDA & FPGA	Yosys, nextpnr, Siemens Catapult HLS, Intel Quartus, Xilinx Vivado, Lattice ECP5, DE1-SoC
Formal & Compilers	CBMC, Spot, SMT Solvers, Temporal Logics, LLVM (IR analysis)
Machine Learning	PyTorch, Scikit-learn, Pandas, NumPy
Infrastructure	Docker, FastAPI, Redis, nginx, AWS Fargate, DigitalOcean, Tailscale, React, Git
Spoken	English, Portuguese